

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (3 Questions)

Course Code:	DSA0607	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	30 (3 Questions × 10 Marks)
CO5 Statement:	<i>Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools (BL6)</i>		
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Section / Batch:	2025/D	Date:	14-08-2026

Code of Conduct

I V.SURYA SATVIKA(192524254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.S.SATVIKA

Instructions

- This test contains 3 questions, each carrying 10 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Visualizing Uncertainty	Probability visualization, confidence intervals, uncertainty interpretation	1	1
Handling Overplotting	Overplotting, transparency, jittering, alternative plots	2	1
Principles of Effective Visualization & Accessibility	Proportional ink principle, misleading axes, accessible color design	3	1
GRAND TOTAL:		30 Marks	

Annexure A – Scenario Based Assessment

1. A weather forecasting agency predicts rainfall probability (%) for a week along with uncertainty:

Day	Rainfall Probability	Confidence Interval
Mon	60	10
Tue	65	20
Wed	70	5
Thu	55	15
Fri	80	25

As a data analyst, design a visualization that clearly communicates both probability and uncertainty. Which day's prediction is most reliable? Justify your answer using visualization principles.

1. Rainfall Probability – Visualization with Uncertainty Suitable

visualization: Error bar chart

- X-axis → Days
- Y-axis → Rainfall Probability (%)
- Point/bar → Predicted rainfall probability
- Error bar → Confidence interval ($\pm$ uncertainty)

Day	Probability (%)	Uncertainty ($\pm$)	Range
Mon	60	10	50–70
Tue	65	20	45–85
Wed	70	5	65–75
Thu	55	15	40–70
Fri	80	25	55–105*

*If this represents a probability bounded between 0% and 100%, the displayed interval should be clipped at 100%.

Interpretation:

- Friday has the highest predicted rainfall probability (**80%**), but its uncertainty is also the highest (**$\pm 25\%$**).

- Wednesday has a prediction of **70% $\pm$ 5%**, giving the narrowest interval.
- Therefore, **Wednesday's prediction is the most reliable/precise.**

Visualization principle: An error-bar chart is suitable because it simultaneously shows the central estimate and its uncertainty. A **smaller error bar means greater precision**. The graph should use a consistent **0–100% Y-axis**, since probability cannot exceed 100%.

2. A mobile app company collects user ratings (1–5) from 500 users. Many users give the same ratings:

Rating	Frequency
1	50
2	80
3	150
4	140
5	80

If plotted directly, the visualization suffers from overplotting. Propose a suitable visualization technique and explain how transparency or jittering can improve clarity. Suggest an alternative plot if needed.

2. Mobile App Ratings – Avoiding Overplotting

Suitable visualization: Jitter plot with transparency or a dot plot.

Since 500 users provide ratings from only 1 to 5, many observations have exactly the same x-value.

Plotting all observations directly would cause points to overlap.

Jittering

Jittering adds a small random horizontal or vertical displacement to overlapping points. This allows individual observations to become visible rather than appearing as one large point.

Transparency

Using semi-transparent points allows overlapping observations to appear darker or denser. Therefore, areas with many observations can be identified easily.

Rating distribution:

Rating	Frequency
1	50
2	80
3	150

4	140
5	80

Interpretation:

- Rating 3 has the highest frequency: 150 users.
- Rating 4 is next with 140 users.
- Ratings 2 and 5 each have 80 users.
- Rating 1 has the lowest frequency: 50 users.

Alternative plot

A bar chart is actually the clearest choice for this summarized frequency data.

- X-axis → Rating (1–5)
- Y-axis → Number of users
- Bar height → Frequency

Conclusion

Use a jittered transparent dot plot when individual user observations need to be shown. If only the given frequency data is available, a bar chart is more appropriate and completely avoids overplotting.

3. A company presents employee satisfaction scores:

Department	Score (%)
HR	70
Sales	75
IT	78
Finance	80

The chart uses a truncated y-axis (starting at 65) and red-green color coding. Identify the issues in this visualization. Redesign it using proper axis design, proportional ink principle, and accessible color choices. Explain how your design avoids misleading interpretation.

1. Truncated Y-axis

The Y-axis starts at 65% instead of 0%.

Because the scores range only from 70% to 80%, starting at 65% can make relatively small differences appear much larger than they really are.

For example, the difference between HR (70%) and Finance (80%) is only 10 percentage points, but a truncated axis can visually exaggerate this difference.

2. Red-green color coding

Using red and green is problematic because many people with color-vision deficiencies have difficulty distinguishing these colors.

Also, color should not be the only method used to communicate information.

Redesigned visualization

Use a simple bar chart:

- X-axis → Department
- Y-axis → Satisfaction Score (%)
- Y-axis starts at 0% and ends at around 100%.
- Use one accessible color for all bars, or use clearly distinguishable colors that do not depend on red vs. green.
- Display the exact percentage above each bar.

Department	Satisfaction
HR	70%
Sales	75%
IT	78%
Finance	80%

Proportional ink principle

The amount of visual area used should correspond fairly to the actual data values. Starting the Y-axis at 0% ensures that the bar lengths are proportional to the satisfaction scores.

Interpretation

Finance has the highest satisfaction score (80%), followed by IT (78%), Sales (75%), and HR (70%).

The redesigned graph shows that the differences are relatively small, rather than visually exaggerating them.

Conclusion:

A properly scaled 0–100% axis, simple accessible colors, and numerical labels make the visualization accurate, proportional, and accessible.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____